

EDP ECOSYSTEM DEEP-DIVE

MODULE 10: UNDERGROUND FORUMS AND DARK WEB MARKETS

Module Number	10
Module Name	Underground Forums and Dark Web Markets
EDP Node Reference	Node 07 (primary): Underground Forum Trust Infrastructure; cross-linkage Nodes 04, 08, 10
Ecosystem Layer	Market Infrastructure / Trust Governance
Upstream Connections	BPH (Module 09 / Node 03); Mixing Services (Module 11 / Node 08); Stealer-Log Markets (Module 01 / Node 10); Crypters (Module 03 / Node 11)
Downstream Connections	IAB Markets (Module 05 / Node 04); Ransomware and RaaS Operations (Module 07); Leak Site Operations (Module 08 / Node 06); Credential Stuffing and Fraud Chains
Research Date	April 2026
Primary Researcher	Reno
Source Tools Used	Perplexity AI; Outpost24; Intel471; Bitsight; Cyjax; SOCRadar; SLCyber; ASEC; Europol
Handling	INTERAGENCY

SECTION 1: WHAT IT IS

Definition and Ecosystem Role

Underground forums and dark-web markets constitute the trust and governance layer of the ransomware supply chain. They are not passive repositories of stolen data or tools; they are active operational infrastructure that enables criminal specialization, coordination, and commerce at scale. Without the reputation systems, escrow mechanisms, and arbitration frameworks these platforms provide, high-value transactions between unknown parties — such as IAB access sales to ransomware affiliates — would be prohibitively risky for both sides.

The ecosystem encompasses five distinct platform types, each occupying a different functional niche. Russian-language, invite-gated forums (Exploit, XSS, RAMP) serve as the institutional core: they host IAB access listings, RaaS program recruitment, and specialist service advertisements within tightly governed communities. English-language breach and data-trade forums (BreachForums variants) operate as higher-volume, lower-trust environments trading database dumps, credential sets, and lower-tier access. Multi-category dark-web markets apply Silk-Road-style marketplace logic (escrow, vendor ratings, dispute resolution) to aggregated goods including stealer logs, botnets, forged documents, and RaaS kits. Specialized ransomware communities host RaaS affiliate recruitment threads, DLS tracking, and tooling distribution. Stealer-log and credential shops function as commodity markets for endpoint logs and credentials, providing search-by-domain or service-type functionality that enables targeted IAB campaigns.

How It Functions: Step-by-Step Operational Logic

- Actor onboarding: New participants are vetted through referrals, vouching threads, or escrow-backed trial transactions. On top-tier Russian forums, accounts require invitation and may require deposit of a cryptocurrency bond.

- Listing and discovery: IABs, malware vendors, BPH providers, and other suppliers post structured offers in category-specific sections. Forum search functions and credential shop query APIs enable targeted discovery by access type, victim sector, geography, or service specification.
- Escrow and transaction: High-value transactions are mediated through on-platform escrow, typically denominated in cryptocurrency. Escrow holds funds until the buyer confirms delivery of promised access or goods.
- Arbitration and dispute resolution: Forum admins and designated arbitrators mediate disputes between buyers and sellers. Arbitration records document inter-actor relationships, payment disputes, access quality claims, and exclusivity violations — intelligence-rich artifacts.
- Reputation maintenance: Vouching threads, positive/negative reviews, and ban records create portable reputation signals. Long-running actors accumulate reputation that functions as collateral for future high-value deals.
- IAB-RaaS coordination: Access sellers post corporate network access offers; ransomware affiliates or managers respond privately or publicly. Intel471 documented at least 70 correlations between IAB forum offers and subsequent DLS victim listings, with a median 19-day lag between forum listing and DLS publication.

Business Model

Forum operators monetize through listing fees, escrow commissions (typically 2-5% of transaction value), premium account subscriptions, and arbitration fees. Market operators earn per-transaction escrow commissions plus vendor registration fees. Credential shop operators earn per-record or per-log-bundle sales revenue. High-volume, high-reputation platforms generate substantial recurring revenue that funds infrastructure, administration, and moderation staff.

Platform Variants

[CONFIRMED] Russian invite-gated forums (Exploit, XSS, RAMP): Long-running communities with strict governance, heavy IAB and RaaS presence, and documented FSB-adjacent behavior by top administrators. Exploit has operated since approximately 2005; XSS since approximately 2013. RAMP positions itself as explicitly ransomware-centric.

[CONFIRMED] English-language breach and data-trade forums (BreachForums variants): Higher-volume, semi-open platforms trading database dumps, credential sets, and combo lists. Subject to recurring LE takedowns but reconstitute quickly under new branding — the LeakBase forum was dismantled in a Europol-coordinated action (Operation Leak) in March 2026, with successor activity emerging rapidly.

[CONFIRMED] Multi-category dark-web markets: Tor-hosted marketplaces aggregating diverse criminal goods and services under Silk-Road-style market logic. Aggregate vast credential volumes: Bitsight's 2025 underground report identified 2.9 billion unique credentials and 7.7 million endpoint logs across observed platforms.

[CREDIBLE] Specialized ransomware and extortion communities: Smaller forums and dark-web hubs hosting RaaS program announcements, affiliate vetting, and IAB-ransomware matchmaking. RAMP's extortion section is the primary example; branding mimicry (e.g., DarkVault copying LockBit DLS design) is documented.

[CONFIRMED] Stealer-log and credential shops: Markets optimized for endpoint logs and credentials from stealer malware and data breaches. Search functions by domain, email, or service type enable IABs and credential-stuffing operators to rapidly identify high-value targets.

SECTION 2: KEY ACTORS AND EXAMPLES

Platform Archetypes and Named Examples

Archetype	Known Examples	Primary Function	Goods / Access Traded	Geo Orientation	Confidence
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Russian invite-gated forums	Exploit (est. ~2005), XSS (est. ~2013), RAMP	IAB listings, RaaS recruitment, specialist service ads, governance/arbitration	Corporate network access, malware, BPH services, RaaS affiliate slots	Russian-language; global victim targeting	[CONFIRMED] High
English-language breach forums	BreachForums (multiple iterations), RaidForums (seized 2022), successors post-2026 takedown	Database dump trading, credential resale, lower-tier access brokering	Database dumps, credential sets, combo lists, VPN credentials, small-org access	English-language; global reach	[CONFIRMED] High
Multi-category dark-web markets	Various Tor-hosted markets (no current dominant platform after successive takedowns)	Aggregated goods marketplace with escrow and vendor ratings	Stealer logs, RDP/VPN access, botnets, spam services, forged documents, RaaS kits	Multilingual; global	[CONFIRMED] High
Specialized RaaS communities	RAMP extortion sections; smaller dark-web RaaS hubs; DarkVault, Qilin, Hellcat communities	RaaS program announcements, affiliate vetting, IAB-RaaS matchmaking, DLS trend tracking	RaaS affiliate slots, IAB introductions, operational guidance, tooling	Russian-language primary; some English sections	[CREDIBLE] Moderate-High
Stealer-log and credential shops	Genesis Market successors; various Tor-hosted log shops	Credential and endpoint log sales with search-by-domain/service functionality	Endpoint logs, browser cookies, VPN/SaaS/cloud/webmail credentials, session tokens	Global supply; global buyer base	[CONFIRMED] High

Geographic Concentration and Scale

Russian-language forums dominate the high-value end of the IAB market. In a 2024 Cyjax review of IAB activity targeting the Middle East, 52.85% of IAB listings resided on XSS; Exploit was the second major venue. Outpost24 identifies Exploit, XSS, and RAMP as the three most significant venues for corporate access sales, with documented handle correlations linking IAB sellers to named RaaS programs including Thanos and AvosLocker.

At the aggregate level, Bitsight's 2025 State of the Underground report documents 2.9 billion unique credentials and 7.7 million endpoint logs available across observed underground markets and forums. Ransomware attacks tracked via underground and DLS activity grew approximately 25% year-over-year; ransomware groups publicly listing victims increased approximately 38% in 2024 (SLCyber). The number of active ransomware groups posting victims grew 11% in H1 2024 versus H1 2023.

State Adjacency

[ANALYST INFERENCE] Top-tier Russian forum administrators are assessed with moderate-high confidence to maintain protection relationships with Russian state security services (FSB, potentially SVR). No confirmed open-source attribution existed for Exploit or XSS administrators until the July 2025 arrest of the XSS administrator (operating as Toha) in France; Exploit administration remains unattributed. Operational longevity (15+ and 10+ years respectively without meaningful Russian law enforcement action, the July 2025 XSS action being a Western one), the explicit carve-out of CIS-region victims from acceptable targeting in many RaaS programs hosted on these forums, and the alignment of forum governance with Dark Covenant protection patterns all support this inference. Dark Covenant 3.0 (Recorded Future) screening is required before any attribution action targeting these administrators.

[**CONFIRMED**] RAMP administration has been associated in multiple private-sector reports with active ransomware ecosystem participants. RAMP's explicit focus on ransomware coordination and its role as a venue for RaaS program advertisements creates elevated operational significance relative to general-purpose forums.

SECTION 3: INFRASTRUCTURE DEPENDENCIES

Upstream Dependencies

Bulletproof Hosting (Module 09 / Node 03): Forums and dark-web markets depend on BPH for Tor hidden service hosting, DDoS protection, abuse-complaint resistance, and domain/IP cycling. The hosting stack for top-tier Russian forums is assessed to use BPH providers with FSB-adjacent protection. BPH disruption is the single highest-leverage upstream action.

Cryptocurrency infrastructure (Node 08 / Module 11; Node 01 / Module 12): Forum escrow, market transactions, and vendor payments are denominated in cryptocurrency. Mixing services launder proceeds; OTC brokers and exchanges convert to fiat. Disruption of financial rails degrades actor confidence in escrow systems — a compounding effect on forum trust infrastructure.

Stealer-log supply chain (Node 10 / Module 01): Stealer malware campaigns generate the endpoint logs and credentials that populate credential shop inventory. Without sustained stealer output, credential shop liquidity degrades, reducing the fuel available for IAB targeting and credential-stuffing chains.

Anonymization and operational security infrastructure (Node 15): Forum and market operators, vendors, and buyers rely on VPNs, Tor, and residential proxy networks for operational security. Degradation of anonymization services increases attribution risk for forum participants.

Downstream Outputs

Forums and markets produce four primary outputs that feed downstream ecosystem nodes: (1) corporate network access (IAB offers → ransomware deployment); (2) stolen credentials and logs (→ credential stuffing, ATO fraud, further access brokering); (3) RaaS affiliate recruitment (→ expanded ransomware deployment capacity); and (4) operational intelligence on law enforcement activity, victim-sector trends, and tooling (→ attacker adaptation).

Critical Chokepoints

Chokepoint	Description	Primary Owner	Disruption Method
Forum reputation and escrow infrastructure	Trust layer enabling high-value transactions between unknown parties; collapse of escrow confidence would significantly degrade transaction volume	Forum admins; escrow service providers	Escrow address designation (OFAC); admin compromise or exposure; counter-intelligence trust-degradation operations
Admin and moderator network	Governance actors enforcing rules, conducting arbitration, and maintaining forum stability; admin capture or exit creates governance vacuum	Forum admins; senior trusted members	Attribution + legal action (MEDIUM-HIGH backfire for RU-based admins); infiltration; handle correlation via Dark Covenant screening
BPH hosting stack	Hidden service hosting for Tor-accessible forums and markets; loss of BPH forces migration and increases exposure during transition	BPH operators (Node 03)	Phase A BPH disruption (upstream ISP/registrar action; FVEY IC coordination)
IAB-RaaS	Forum sections and private	Forum moderation;	Persistent infiltration; targeted

coordination channels	threads where access sellers and ransomware buyers match; disrupting these channels extends the IAB-to-deployment lag	FVEY LE infiltration assets	thread/actor removal; counter-operation to degrade buyer-seller trust
Stealer-log shop inventory pipeline	Fresh log supply from active stealer campaigns; inventory staleness degrades shop utility for targeted IAB operations	FVEY LE + private sector (Node 10 / Module 01)	Stealer infrastructure takedowns; C2 disruption; botnet sinkholing (cross-reference Module 02 / Node 05)

Cross-Module Linkages

Module	Node	Linkage Type	Direction	Description
01 Stealers	10	Supply	Upstream	Stealer malware campaigns generate logs sold on credential shops; credential shop inventory is the IAB fuel supply
02 Loaders	05	Supply / Listing	Upstream / Bidirectional	Botnet access and loader services are listed as products on dark-web markets; forum ads recruit loader operators
05 IABs	04	Market Venue	Bidirectional	Forums are the primary venue for IAB offer listings; IAB activity is the primary commercial driver of top-tier RU forum value
07 Ransomware / RaaS	Cross-cutting	Recruitment / Coordination	Bidirectional	RaaS program ads, affiliate recruitment, and IAB-RaaS matchmaking are core forum functions; forums are the primary RaaS recruitment channel
08 Leak Site Ops	06	Dissemination	Downstream	Forum communities track, amplify, and analyze DLS publications; forum threads serve as secondary dissemination for leak-site data
09 BPH	03	Infrastructure	Upstream	BPH provides hosting infrastructure for Tor hidden service forums and markets; loss of BPH forces costly platform migration
11 Crypto Mixers	08	Financial	Upstream	Mixing services process proceeds from forum and market transactions; escrow payouts flow through mixers before cash-out

SECTION 4: DISRUPTION LEVERAGE POINTS

Primary Leverage Points

Lever	Owner	Best Method	Backfire Risk	EDP Phase
Persistent intelligence exploitation of RU-language forums	FVEY IC + private sector (Intel471, Recorded Future, Flashpoint)	Long-term infiltration; IAB handle correlation; arbitration record collection; 19-day window	LOW	Phase B — ongoing

		exploitation for victim early warning		
English-language forum takedown	FVEY LE (FBI, NCA, Europol); coordinated multi-agency	Infrastructure seizure; administrator arrest; coordinated notification to carryover actors; pre-positioned infiltration of successor platform	LOW (reconstitution expected; sequence for max intel yield)	Phase B — opportunistic
Escrow service disruption / designation	Treasury / OFAC; blockchain forensics (Chainalysis, TRM, Elliptic)	OFAC designation of identified escrow cryptocurrency addresses; exposure of escrow provider identities to degrade actor confidence	LOW	Phase B — compounds Phase A
Trust and reputation degradation (counter-intelligence)	FVEY IC; specialized LE units	Fabricated dispute injection; counter-intelligence operations to create actor suspicion; exposure of LE infiltration to trigger self-purge	LOW-MEDIUM	Phase B — advanced, sequenced after forum penetration
RU-language forum admin attribution	FVEY IC; requires Dark Covenant 3.0 pre-screening	Handle correlation; persona linkage; infrastructure attribution; do NOT action without full protection-relationship mapping	MEDIUM-HIGH — FSB adjacency likely; requires senior authorization	Phase B — high-value, high-risk; long-cycle action

Compounding Actions

- Sequence Phase A financial actions (Node 01 OTC, Node 02 exchanges) before Phase B forum pressure. Degrading financial rails reduces actor confidence in escrow systems, amplifying downstream trust-infrastructure effects on forums.
- Exploit the 19-day IAB-to-DLS window: position collection assets on Exploit, XSS, and RAMP to detect IAB listings in real time; correlate with active LE operations for victim early warning and potentially disruptive intervention.
- Coordinate credential shop disruption (Node 10) with stealer C2 takedowns (Module 01 / Node 10 and Module 02 / Node 05) to degrade shop inventory freshness — forcing IABs to work from stale data that increases operational risk.
- Pre-position monitoring infrastructure in English-language forum successor environments before and immediately after takedown; the reconstitution period (weeks to months) is the highest-collection-yield window.
- Share IAB listing intelligence with targeted victim sectors (healthcare, critical infrastructure, legal) as private-sector early warning to enable defensive action within the 19-day window.

SECTION 5: RESILIENCE AND REPLACE DIFFICULTY

Replace Difficulty by Platform Type

Node 07 carries a HIGH replace difficulty rating in the EDP Dependency Map. However, this aggregate rating masks a critical internal distinction: Russian-language institutional forums and English-language transactional forums have fundamentally different durability profiles that require separate disruption assessments.

Platform Type	Replace Difficulty	Key Durability Driver	Est. Recovery Time if Disrupted	Confidence
Russian invite-gated forums (Exploit, XSS, RAMP)	VERY HIGH	10-20 year trust networks; admin governance; FSB-adjacent protection; strict vetting makes replication extremely slow	Years — if disruption is achievable at all	[ANALYST INFERENCE] High confidence on durability; low confidence on any disruption pathway
English-language breach forums	LOW-MEDIUM	Open/semi-open participation; rapid rebranding; broad existing actor base; portable vendor relationships	Weeks to months (RaidForums → BreachForums: weeks; BreachForums v1 → v2: months)	[CONFIRMED] Historical pattern well-documented
Multi-category dark-web markets	MEDIUM	Market infrastructure is reconstitutable; vendor relationships are portable to new platforms; market logic is standardized	2-6 months typical (Silk Road, AlphaBay, Hansa successor patterns)	[CONFIRMED] Well-documented historical reconstitution
Specialized RaaS communities	MEDIUM	Smaller actor base; platform survival tied to associated RaaS program viability; program shutdown accelerates platform decline	Months (program-dependent)	[CREDIBLE] Moderate — limited case studies
Credential and stealer-log shops	LOW	Minimal trust infrastructure required; inventory is portable; technical barriers to launch are low	Days to weeks (Genesis Market successors emerged within days)	[CONFIRMED] Genesis Market post-seizure pattern confirmed

Historical Reconstitution Record

Platform	Disruption Event	Date	Reconstitution	Notes
RaidForums	Europol / DOJ seizure; admin arrested	2022	BreachForums launched within weeks	Near-instant brand migration; same actor ecosystem largely intact
BreachForums v1	FBI arrest of admin (Pompompurin)	2023	BreachForums v2 operational within months	Successor operated by different admin; community largely reconstituted
Genesis Market	FBI / Europol Operation Cookie Monster	April 2023	Multiple successor credential shops within months	Product inventory partially preserved; vendor ecosystem migrated rapidly
Major stolen-data forum (unnamed)	Europol-led takedown	2026	Successor activity emerging; monitoring ongoing	Consistent with prior English-language forum reconstitution pattern

Exploit / XSS	XSS seized July 2025 (admin arrested in France); Exploit no LE action to date	N/A	N/A — continuous operation	[ANALYST INFERENCE] Exploit 15+ yr continuity and undisrupted; XSS seized July 2025 (admin arrested) with roughly two thirds of users migrating to DamageLib within weeks. These forums are not impervious; the disruption pathway is breaking forum trust, not seizing infrastructure alone
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Redundancy and Ecosystem Adaptation

The underground forum ecosystem demonstrates high systemic redundancy. At any given time, multiple platforms exist across the full spectrum — invite-gated RU forums, English-language breach boards, dark-web markets, and specialized communities — each capable of absorbing displaced actors from disrupted platforms. High-value actors on Russian forums maintain relationships across multiple platforms and can rapidly redirect activity if one venue is disrupted or compromised.

Private encrypted channels (Telegram groups, encrypted messaging) increasingly serve as fallback coordination venues when forum activity is disrupted. This trend, documented by multiple private-sector researchers, represents a structural adaptation that reduces dependence on any single forum platform — though at the cost of the governance and escrow features that forums provide.

SECTION 6: INDICATORS AND KPIS

Ecosystem Health Indicators

Indicator	Normal State (2024-2025 Baseline)	Under Pressure / Degraded
IAB listing volume on Exploit / XSS / RAMP	Active daily listings; multiple corporate access offers per week per forum; documented XSS dominance (52.85% of IAB listings per Cyjax 2024)	Reduced listing frequency; shorter listing durations; increased listing failures; migration to private channel offers
IAB-to-DLS correlation lag	Median 19 days from IAB forum listing to DLS victim publication (Intel471, June 2024 - May 2025, 70-case dataset)	Lag extension to 45+ days indicates disrupted buyer-seller matching; correlation collapse indicates IAB channel failure
Forum arbitration and escrow activity	Regular dispute resolution; consistent arbitration outcomes; low escrow abandonment rate	Arbitration backlogs; unresolved disputes; escrow abandonment; admin inactivity signals governance breakdown
Credential shop inventory volume and freshness	Millions of fresh logs per month; domain-searchable inventory; 2.9B unique credentials across observed platforms (Bitsight 2025)	Reduced inventory freshness; older log batches dominating; price compression; shop closures
RaaS recruitment thread	Active program ads with affiliate	Reduced recruitment activity; affiliate

activity on forums	vetting; vouching threads; repeat relationship documentation	defection signals; ghost program ads; recruitment migration to private channels
Underground data breach discussion volume	Up 43% YoY per Bitsight 2025; endpoint logs for sale up 13%; compromised credentials up 34% vs 2023	Year-over-year decline in discussion volume; reduced posting frequency; platform migration patterns visible

Disruption KPIs

KPI	Baseline (2024-2025)	Target Under Disruption	Collection Method
IAB listings per month on Exploit / XSS / RAMP combined	Est. 200-400 tracked per month (Outpost24, Intel471 combined reporting)	Below 50% of baseline; persistent decline over 90-day window	Persistent forum monitoring; HUMINT; private-sector feed correlation
IAB-to-DLS correlation lag (median, days)	19 days (Intel471 70-case dataset, June 2024 - May 2025)	Greater than 45 days; or correlation breakdown for 3+ consecutive months	Intel471 / Flashpoint / Recorded Future IAB tracking; DLS monitoring cross-correlation
Fresh unique credentials available for sale (quarterly)	~725M credentials per quarter (annualizing Bitsight 2.9B figure)	Greater than 25% reduction in fresh credentials over two consecutive quarters	Bitsight; DarkOwl; SpyCloud underground data feeds
Ransomware groups actively posting victims (monthly count)	38% growth in active groups in 2024; H1 2024 up 11% vs H1 2023 (SLCyber)	Year-over-year decline in active group count; 20%+ reduction sustained over two quarters	DLS monitoring (Ransomlooker, RansomWatch); SOCRadar; SLCyber quarterly reports
Time-to-reconstitution for English-language forum post-takedown	Weeks to months historically	N/A as a target (reconstitution is expected); metric is pre-positioned monitoring activation within 72 hours of successor detection	Forum seeding; actor handle tracking across platforms; Tor onion scanning

Alert Thresholds

Threshold	Trigger Condition	Response
Exploit / XSS admin attribution confidence exceeds 70%	Dark Covenant 3.0 screening complete; admin handle correlated to real-world identity; FSB protection relationship mapped or ruled out	Escalate to FVEY IC senior review before any action; do not proceed to public attribution without full protection-relationship mapping and senior authorization
New English-language forum launched post-takedown within 30 days	Successor platform detected with carryover actor handles or inventory	Activate pre-positioned monitoring; do not re-target immediately; assess intel yield vs. disruption value before follow-on action
IAB-DLS correlation spike: 50+ new correlations in 30-day window	Intel471 or equivalent reports abnormal IAB listing volume correlating to DLS publications	Cross-reference with active LE operations; assess for coordinated campaign; activate victim early-warning dissemination to at-

		risk sectors
RAMP admin change or platform restructuring	New admin announcement; major TOS change; RAMP migration to new infrastructure	Assess for LE infiltration signal, rival actor takeover, or FSB-directed restructuring; Dark Covenant rescreen required
Credential volumes spike greater than 50% YoY	Bitsight or equivalent reports aggregate credential availability exceeding 4.5B unique credentials	Correlate with Module 01 (Stealers) and Module 02 (Loaders) activity surge; assess for coordinated stealer campaign targeting specific victim sectors

SECTION 7: SOURCES AND CONFIDENCE

Primary Sources

Intelligence and Private Sector Threat Research:

- Outpost24 — "Demystifying Initial Access Brokers (IABs) and their links to ransomware": Exploit/XSS/RAMP analysis; IAB-RaaS forum overlaps; arbitration examples; vouching thread documentation (Thanos, AvosLocker operators).
- Intel471 — "How initial access offers power intrusions and ransomware": 70-case IAB-ransomware correlation dataset; 19-day median delay; IAB-to-DLS matching methodology.
- Cyjax — "2024 Year in Review: ransomware groups, hacktivists, and IABs targeting the Middle East": Forum prominence data; XSS 52.85% IAB listing share; Exploit second venue.
- Bitsight — "2025 State of the Underground" report: 2.9B unique credentials; 7.7M endpoint logs; ransomware attack growth ~25%; breach discussions up 43%; endpoint logs up 13%; credentials up 34%.
- SOCRadar — "Annual Dark Web Report 2024": 1,043,781 email-password credentials from top-visited domains; 2,080,000+ dark-web exposure alerts (15.4% YoY increase); LockBit 3.0 at 14.63%; RansomHub at 9.03%; Play at 6.04%.
- SLCyber — "Ransomware in H1 2024: Trends from the Dark Web" and "Most Prolific Ransomware Groups": 11% rise in listed ransomware victims vs H1 2023; 38% increase in number of groups posting victims; DarkVault, APT73, Quilong, Hellcat tracking.
- ASEC — "Ransomware Groups and Cybercrime Forums and Markets in June 2024": DarkVault LockBit-clone branding; forum and DLS trend tracking.
- CRIF — "Cyber attacks on the rise: data theft on the dark web up 15% in 2024": VPN services 34.3% of exposed accounts; social networks 23.9%; public-sector accounts doubled YoY.

Law Enforcement:

- Europol — RaidForums takedown (2022); BreachForums actions; major stolen-data forum dismantling (2026).
- FBI — Genesis Market (Operation Cookie Monster, April 2023); BreachForums v1 (2023).
- DOJ — RaidForums administrator extradition and prosecution (2022-2023).

Analytical Framework:

- Recorded Future — Dark Covenant 3.0: FSB protection reflex framework applied to forum admin attribution assessment.

Confidence Assessment by Topic

Topic	Confidence Level	Basis	Key Limitations
Forum roles and IAB-RaaS	[CONFIRME	Multiple independent private-	Private-sector access to

operational linkages	D] CONFIRMED	sector reports; Outpost24, Intel471, Cyjax with separate research methodologies	forum data varies; some reporting based on partial forum visibility
19-day median IAB-to-DLS lag	[CREDIBLE] CREDIBLE	Single primary source (Intel471); 70-case dataset; June 2024 - May 2025 window; directionally consistent with other reporting	Methodology not fully publicly disclosed; dataset may not be representative across all RaaS families or geographies
Underground market aggregate scale (2.9B credentials, 7.7M logs)	[CREDIBLE] CREDIBLE	Bitsight 2025 cross-platform aggregation; consistent with other market-monitoring sources	Cross-platform deduplication methodology not fully disclosed; figures likely conservative given forum access limitations
XSS 52.85% IAB listing share	[CREDIBLE] CREDIBLE	Cyjax 2024 Middle East-focused review; single-region, single-period dataset	Regional focus limits global applicability; other geographies may show different forum distributions
FSB adjacency of top-tier RU forum admins	[ANALYST INFERENCE] ANALYST INFERENCE	No confirmed open-source attribution; inferred from operational longevity, CIS-victim carve-outs in hosted RaaS programs, and Dark Covenant 3.0 framework	Absence of LE action may reflect operational security rather than protection; cannot distinguish between FSB protection and superior OPSEC
English-language forum reconstitution timeline	[CONFIRMED] D] CONFIRMED	Multiple documented cases: RaidForums → BreachForums, BreachForums v1 → v2, Genesis Market successors	Each reconstitution event is unique; specific timelines vary; successor quality and trust may differ from predecessor

Intelligence Gaps

- Exploit administrator identities: no confirmed open-source attribution, the highest-priority gap for an Exploit disruption action. XSS: its administrator was arrested in France in July 2025 and is identified, so this gap is now closed for XSS.
- RAMP administrator identities and FSB/GRU adjacency: RAMP's ransomware-centric function elevates the operational significance of this gap.
- Escrow service provider identities: The specific cryptocurrency escrow services used by top-tier forums are not publicly documented. These represent potential OFAC designation targets.
- IAB-DLS correlation methodology: Intel471's 70-case dataset is the only publicly available correlation study. Independent validation is needed to confirm the 19-day lag figure and expand to additional ransomware families.
- Forum penetration depth: The degree to which FVEY and partner intelligence services currently have persistent access to Exploit, XSS, and RAMP is not reflected in open-source reporting.

SECTION 8: ANALYST ASSESSMENT

Key Takeaway

Underground forums are the institutional backbone of the ransomware supply chain, not peripheral infrastructure. Russian-language forums — particularly Exploit and XSS — function as multi-decade criminal institutions with governance frameworks, dispute resolution systems, and reputation infrastructure that far exceeds the

sophistication of any single ransomware group or IAB operation. The July 2025 seizure of XSS (admin arrested in France) shows even these institutions are reachable; what proved durable was the trust and governance layer, and that is also what collapsed fastest once the administrator was compromised, with roughly two thirds of users migrating to a successor within weeks. The 19-day median lag from IAB forum listing to DLS victim publication (Intel471) is the most operationally significant metric in this module: it confirms that forum-based IAB-RaaS coordination is a measurable, trackable, and potentially exploitable precursor indicator of imminent ransomware attacks.

The Node 07 HIGH replace difficulty rating is accurate at the aggregate level but obscures a critical distinction: Russian-language institutional forums are assessed at VERY HIGH replace difficulty; the July 2025 XSS takedown (a French and Ukrainian action) shows they are not impervious, though the decisive damage came from the resulting trust collapse and user migration rather than the seizure itself, so disruption depends on breaking forum trust, not just seizing infrastructure. English-language forums are LOW-MEDIUM replace difficulty and are appropriate targets for LE action, but only when sequenced to maximize pre-action intelligence yield and followed by persistent infiltration of successor platforms.

Priority Recommendation

Immediate: Optimize collection architecture for the 19-day IAB-to-DLS window. Collection assets on Exploit, XSS, and RAMP should be calibrated to detect, correlate, and disseminate IAB listing intelligence within 48-72 hours of posting. This window represents the highest-leverage point for victim early warning and potentially disruptive intervention — it is actionable with existing collection infrastructure and does not require any disruptive forum action.

Near-term: Initiate Dark Covenant 3.0 screening for Exploit and XSS administrator handle clusters. The intelligence gap on admin identities and their FSB protection relationships is the single largest constraint on any escalated forum action. Screening should be completed before any attribution product is developed or disseminated.

Medium-term: Map escrow service providers used by top-tier forums as potential OFAC designation targets. Escrow designation is a LOW-backfire, Phase B compounding action that degrades forum transaction confidence without triggering the FSB protection reflex that individual administrator attribution would risk.

Sequencing note: Phase A financial actions (Nodes 01, 02 — OTC brokers, exchanges) must precede Phase B forum pressure actions. Degrading financial rails reduces actor confidence in escrow systems, amplifying the destabilizing effect of any subsequent forum trust-infrastructure action. The playbook sequence is load-bearing here.

Connection to EDP Disruption Playbook

Node 07 (Underground Forum Trust Infrastructure) sits in Phase B alongside Nodes 04 (IAB Markets) and 08 (Mixing Services). These three nodes constitute the market and trust infrastructure layer of the ransomware supply chain — the layer that enables criminal specialization and commerce at scale. Phase B actions are most effective when Phase A financial disruption (Nodes 01, 02, 03) has already degraded actor confidence in financial infrastructure.

Within Phase B, forum disruption compounds IAB market pressure (Node 04): degrading the forums degrades the primary venue for IAB offers, compressing the supply of access available to ransomware affiliates. Simultaneously, escrow disruption compounds mixer and financial rail pressure (Node 08). The three Phase B nodes are mutually reinforcing — coordinated simultaneous pressure across all three maximizes disruption effect.

Dependency Map Update Recommendations

Current Node 07 Field	Current Value	Proposed Change	Rationale
Replace Difficulty	HIGH (single rating)	Sub-categorize: RU-language forums = VERY HIGH; EN-language forums = LOW-MEDIUM; credential	Fundamentally different durability profiles require separate disruption assessments and resource allocation. Treating as single

		shops = LOW	category understates RU-forum resilience and overstates EN-forum resilience.
Backfire Risk	LOW	Adjust: RU-language forum admin attribution = MEDIUM-HIGH; escrow/infrastructure actions = LOW; EN-language takedowns = LOW	Admin attribution for RU-language forums carries FSB adjacency risk per Dark Covenant 3.0 framework. Aggregate LOW rating creates false confidence for any admin-targeting action.
No sub-indicator for IAB-RaaS coordination channels	N/A	Add IAB-forum coordination as a tracked sub-indicator under Node 07 with dedicated KPI (IAB-to-DLS lag time)	The 19-day correlation window is the most operationally actionable metric in the ransomware supply chain. It warrants dedicated tracking and should be a standing requirement for collection architecture.
Primary Owner	FVEY LE + private sector underground monitoring	Add: FVEY IC (covert collection) as co-primary for RU-language forums; LE as primary only for EN-language forums and markets	LE-led action against RU-language forums is not viable; IC collection is the primary instrument. Distinguishing ownership by platform type enables more realistic disruption planning.

Follow-On Research

- Dark Covenant 3.0 screening for Exploit, XSS, and RAMP administrator handle clusters: prioritize FSB and MVD protection relationship mapping. Required before any attribution product or escalated forum action.
- Escrow service provider mapping: identify cryptocurrency addresses and third-party escrow services used by top-tier forums; assess each for OFAC designation viability.
- RAMP administrator attribution: cross-reference admin handles with known RaaS operator personas; assess GRU or FSB adjacency given RAMP's explicit ransomware coordination function.
- IAB-to-DLS correlation dataset expansion: cross-validate Intel471's 70-case dataset with independent data from Recorded Future, Flashpoint, and Mandiant. Extend analysis to additional ransomware families beyond Thanos and AvosLocker cases.
- English-language forum successor pre-positioning: develop monitoring infrastructure for post-takedown successor platforms before the next major EN-language forum action to ensure immediate collection capability in the reconstitution window.
- Telegram and private-channel migration tracking: quantify the degree to which high-value actors are migrating IAB-RaaS coordination from forums to private encrypted channels, and assess whether this represents a structural shift that degrades the operational utility of forum monitoring.